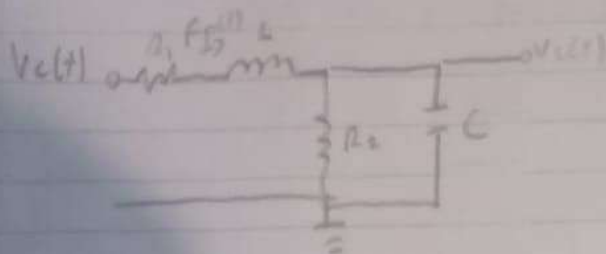


1. Dibuja el diagrama del sistema cardiovascular



$$2. V_c(t) = R_1 i_1(t) L \frac{di_1(t)}{dt} + R_2 [i_1(t) - i_2(t)]$$

$$R_2 [i_1(t) - i_2(t)] = \frac{1}{C} \int i_2(t) dt$$

$$v(t) = \frac{1}{C} \int i_2(t) dt$$

Transformada

$$V_c(s) = R_1 I_1(s) + L s I_1(s) + R_2 [I_1(s) - I_2(s)]$$

$$R_2 [I_1(s) - I_2(s)] = \frac{I_2(s)}{sC}$$

$$V_c(s) = \frac{I_2(s)}{sC}$$

$$R_2 I_1(s) - R_2 I_2(s) = \frac{I_2(s)}{sC}$$

$$R_2 I_1(s) = \frac{I_2(s)}{sC} + R_2 I_2(s)$$

$$R_2 I_1(s) = \left(\frac{1}{sC} + R_2 \right) I_2(s)$$

$$I_1(s) = \left(\frac{1}{sC} + R_2 \right) I_2(s) / R_2$$

$$I_1(s) = \left(\frac{1}{sCR_2} + 1 \right) I_2(s)$$

$$I_1(s) = \left(\frac{1}{sCR_2} + \frac{R_2}{R_2} \right) I_2(s)$$

$$I_1(s) = \left(\frac{CR_2 s + 1}{CR_2 s} \right) I_2(s)$$

$$V_c(s) = R_1 \left(\frac{CR_2 s + 1}{CR_2 s} \right) I_2(s) + L \left(\frac{CR_2 s + 1}{CR_2 s} \right) I_2(s) + R_2 \left(\frac{CR_2 s + 1}{CR_2 s} \right) I_2(s) - R_2 I_2(s)$$

$$V_c(s) = \left[\frac{R_1 CR_2 s + R_1}{CR_2 s} + \frac{L CR_2 s + L}{CR_2 s} + \frac{R_2 CR_2 s + R_2}{CR_2 s} - R_2 \right] I_2(s)$$

$$V_c(s) = \left[\frac{L R_1 CR_2 s^2 + (R_1 CR_2 + L R_2 + R_1 CR_2 - R_2 CR_2) s + R_1 + R_2 CR_2}{CR_2 s} \right] I_2(s)$$

$$V_c(s) = \left[\frac{L R_1 CR_2 s^2 + (R_1 CR_2 + L R_2 + R_1 CR_2 - R_2 CR_2) s + R_1 + R_2 CR_2}{CR_2 s} \right] I_2(s)$$

$$V_c(s) = \left[\frac{L R_1 CR_2 s^2 + (R_1 CR_2 + L R_2 + R_1 CR_2 - R_2 CR_2) s + R_1 + R_2 CR_2}{CR_2 s} \right] I_2(s)$$

$$\frac{V_c(s)}{V_c(s)} = \left[\frac{L R_1 CR_2 s^2 + (R_1 CR_2 + L R_2 + R_1 CR_2 - R_2 CR_2) s + R_1 + R_2 CR_2}{CR_2 s} \right] I_2(s)$$

$$\frac{V_c(s)}{V_c(s)} = \frac{R_2 s}{s^2 L R_2 + (R_1 R_2 L)s + (R_1 + R_2)}$$

$$\frac{V_c(s)}{V_c(s)} = \frac{R_2}{(L R_2)s^2 + (R_1 R_2 L)s + R_1 + R_2}$$

Modelo de estado

$$R_2 [\dot{i}_1(t) - \dot{i}_2(t)] = \frac{1}{L} \int \dot{i}_2(t) dt$$

$$\dot{i}_1(t) - \dot{i}_2(t) = \left(\frac{1}{L} \int \dot{i}_2(t) dt \right) \frac{1}{R_2}$$

$$\dot{i}_1(t) = \left(\frac{1}{L} \int \dot{i}_2(t) dt \right) \frac{1}{R_2} + \dot{i}_2(t)$$

II

$$R_1 \dot{i}_1(t) + L \frac{d\dot{i}_1(t)}{dt} + R_2 [\dot{i}_1(t) - \dot{i}_2(t)] = V_c(t)$$

$$R_2 [\dot{i}_1(t) - \dot{i}_2(t)] = V_c(t) - R_1 \dot{i}_1(t) - L \frac{d\dot{i}_1(t)}{dt}$$

$$\dot{i}_1(t) - \dot{i}_2(t) = [V_c(t) - R_1 \dot{i}_1(t) - L \frac{d\dot{i}_1(t)}{dt}] \frac{1}{R_2}$$

$$-\dot{i}_2(t) = [V_c(t) - R_1 \dot{i}_1(t) - L \frac{d\dot{i}_1(t)}{dt}] \frac{1}{R_2} - \dot{i}_1(t)$$

$$\dot{i}_2(t) = \dot{i}_1(t) - [V_c(t) - R_1 \dot{i}_1(t) - L \frac{d\dot{i}_1(t)}{dt}] \frac{1}{R_2}$$

$$\dot{i}_1(t) = \left[\frac{1}{L} \int \dot{i}_2(t) dt \right] \frac{1}{R_2} + \dot{i}_2(t)$$

$$\dot{i}_2(t) = \dot{i}_1(t) - [V_c(t) - R_1 \dot{i}_1(t) - L \frac{d\dot{i}_1(t)}{dt}] \frac{1}{R_2}$$

$$V_s(t) = \frac{1}{L} \int \dot{i}_2(t) dt$$